Variance of Discrete random variable

Definition: In a probability distribution Variance is the average of sum of squares of deviations from the mean. The variance of the random variable X can be defined as.

$$\text{Var}(X) = E(X - E(X))^2$$

$$= E(X^2) - (E(X))^2$$

Example 9.10

When a die is thrown X denotes the number turns up. Find $E(X)$, $E(X^2)$ and $\text{Var}(X)$.

Solution:

Let X denoted that number turns up in a die.

X takes the values 1, 2, 3, 4, 5, 6 with probabilities $\frac{1}{6}$ for each.

Therefore the probability distribution is

X	1	2	3	4	5	6
P(x)	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$E(X) = \Sigma x_i p_i$$

$$E(X) = x_1 p_1 + x_2 p_2 + \ldots + x_6 p_6$$

$$E(X) = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + \ldots + 6 \times \frac{1}{6}$$

$$= \frac{1}{6}(1+2+3+4+5+6)$$

$$= \frac{7}{2}$$

$$E(X^2) = \Sigma x_i^{\,2} p_i$$

$$E(X^2) = x_1^{\,2} p_1 + x_2^{\,2} p_2 + \ldots + x_6^{\,2} p_6$$

$$E(X^2) = \left(1^{2\times}\frac{1}{6}\right) + \left(2^{2\times}\frac{1}{6}\right) + \ldots \left(6^{2\times}\frac{1}{6}\right)$$

$$= \frac{1}{6}(1+4+9+16+25+36)$$

$$= \frac{1}{6}(91) = \frac{91}{6}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$= \frac{91}{6} - \left(\frac{7}{2}\right)^2$$